

LESSON 1.3

Representing a Categorical Variable with Tables

AP Statistics · Unit 1: Exploring One-Variable Data

Warm-Up

1. Look at the list of 20 students' favorite subjects on your warm-up sheet. Can you immediately tell which subject is most popular? Why or why not?
2. Tally each subject and complete the frequency column. Does your total equal 20?
3. From Lesson 1.2: what *type* of variable is “favorite subject”? Is it meaningful to compute the average?
4. What question do you think today's lesson will answer?

We'll revisit your frequency table at the end of the lesson to add one more column.

PRIMARY OBJECTIVE

Students will construct and interpret **frequency** and **relative frequency** tables for a categorical variable. Students will explain why relative frequency — not raw count — is required to compare groups of different sizes (Big Idea: UNC).

By the end of this lesson, I will be able to:

- ▶ Construct a frequency and relative frequency table from raw categorical data.
- ▶ Interpret relative frequencies in context using complete sentences.
- ▶ Explain why relative frequency is necessary for fair comparisons.

Hook — Which School Has More Walkers?

Two schools survey students about how they get to school. Which school has more walkers?

Mode	School A	School B
Bus	120	30
Car	80	50
Walk	40	120
Total	240	200

Hook — The Answer Changes with Relative Frequency

Mode	Sch. A	Rel. Freq. A	Sch. B	Rel. Freq. B
Bus	120	50%	30	15%
Car	80	33%	50	25%
Walk	40	17%	120	60%

School A has more walkers by *count* (40 vs. 30).

School B has far more walkers by *proportion* (60% vs. 17%).

Lesson: Use relative frequency to make fair comparisons.

Vocabulary

Frequency The count of individuals in a particular category.

Relative frequency Frequency $\div$ total; expressed as a proportion (decimal) or percent.

Frequency table A table listing each category of a categorical variable and its count.

Relative frequency table A table listing each category and its relative frequency.

Distribution (categorical) The set of possible values and their associated relative frequencies.

Key check: Relative frequencies must sum to 1 (or 100%).

Building a Relative Frequency Table — Four Steps

1. **List all categories** in the first column.
2. **Tally** the raw data and record the **frequency** (count).
3. **Divide** each count by n (the total) to get the **relative frequency**.
4. **Check:** all relative frequencies must sum to exactly 1.

$$\text{Relative frequency} = \frac{\text{frequency}}{n}$$

Express as a decimal *or* a percent — both are correct on the AP exam.

Worked Example — Pet Ownership ($n = 40$)

Pet type	Frequency	Rel. Frequency
Dog	12	$12/40 = 0.30$
Cat	8	$8/40 = 0.20$
Fish	6	$6/40 = 0.15$
Bird	4	$4/40 = 0.10$
None	10	$10/40 = 0.25$
Total	40	1.00

AP-style interpretation:

“Of the 40 households surveyed, the most common pet type was dog, owned by 30% of households. The least common was bird, owned by 10%.”

Describing a Distribution — AP Language

Every description must include:

- ▶ The **total** n (“of the 40 households surveyed...”)
- ▶ The **variable** in context (“pet type”)
- ▶ The **most** and **least** common categories
- ▶ The **relative frequency** for at least the key categories

Model sentence frame:

“Of the _____ [individuals] surveyed, the most common [variable] was _____, with _____% of responses. The least common was _____ with _____%.”

On the AP exam, writing “30%” without context earns **no credit**. Always anchor your numbers.

Group Activity — Favorite Sport Survey

Part 1 (8 min): Raw data for 30 middle school students.

1. Tally the data and build a complete frequency and relative frequency table.
2. Verify: total = 30, sum of relative frequencies = 1.
3. Write 2–3 sentences describing the distribution.

Part 2 (5 min): A second dataset has $n = 50$.

1. Complete the relative frequency column for the high school data.
2. Compare the two groups. Which sport is more popular proportionally?
3. Can you compare using counts alone? Why or why not?

Exit Ticket — 5 Minutes

A survey of **50 students** asks: "Where do you prefer to study?"

Location	Count	Rel. Freq.
Library	18	?
Home	22	?
Café	6	?
Other	4	?
Total	50	?

1. Complete the relative frequency column.
2. Write one sentence describing the most common location *in context*.
3. What percentage of students study somewhere other than home?

Big Ideas & Preview

Today's core takeaway:

Use **relative frequency** when comparing groups of different sizes. Counts alone are misleading.

Connections:

- ▶ **Lesson 1.4:** The same distributions shown as bar charts and pie charts — tables vs. graphs, trade-offs.
- ▶ **Unit 2:** Two-way tables extend today's one-way tables to two categorical variables.
- ▶ **Unit 4:** Relative frequency = experimental probability — the arithmetic is identical.

Complete your homework (Lesson 1.3 HW) before next class.